

2. Applications of graph colorings

(1) Scheduling Exams

How can the exams at a university be scheduled so that no student has two exams at the same time?

Solution:

This scheduling problem can be solved using a graph model, with vertices representing courses and with an edge between two vertices if there is a common student in the courses they represent.

Each time slot for a final exam is represented by a different color.

A scheduling of the exams corresponds to a coloring of the associated graph.



For example, Suppose we want to schedule some final exams for CS courses with following code numbers:

1007, 3137, 3157, 3203, 3261, 4115, 4118, 4156

Suppose also that there are no common students in the following pairs of courses because of prerequisites:

1007-3137

1007-3157, 3137-3157

1007-3203

1007-3261, 3137-3261, 3203-3261

1007-4115, 3137-4115, 3203-4115, 3261-4115

1007-4118, 3137-4118

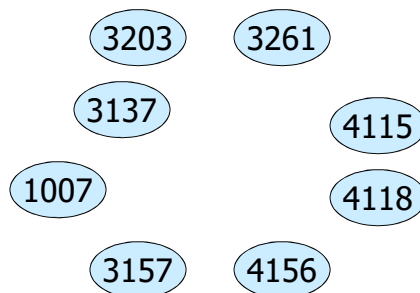
1007-4156, 3137-4156, 3157-4156

How many exam slots are necessary to schedule exams?



Turn this into a graph coloring problem.

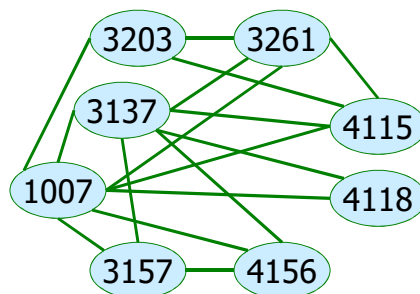
-- Vertices are courses, and edges connect courses which *cannot* be scheduled simultaneously because of possible common students in both courses



3



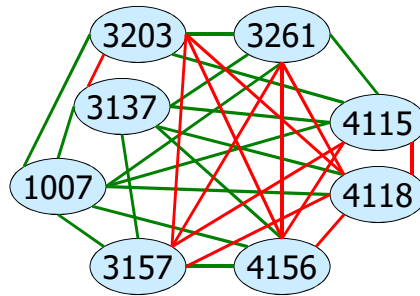
One way to do this is to put edges down where students mutually excluded...



4



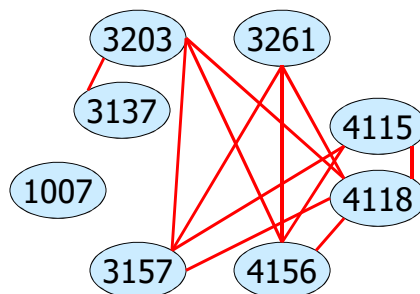
...and then compute the complementary graph:



5



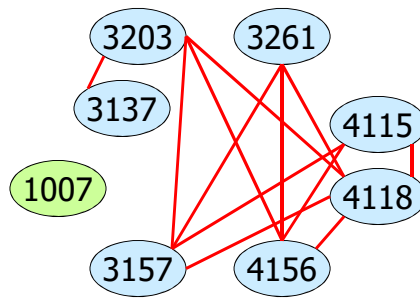
...and then compute the complementary graph:



6



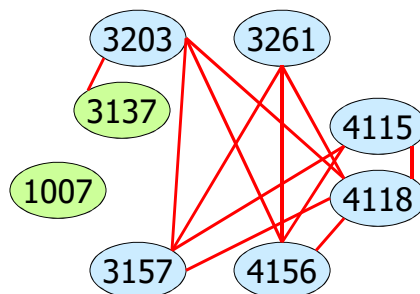
Coloring ...



7



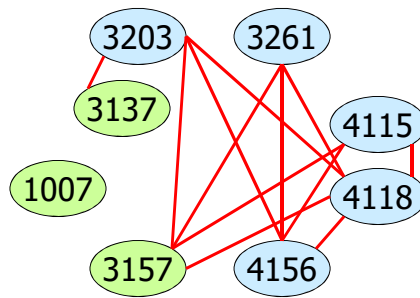
Coloring ...



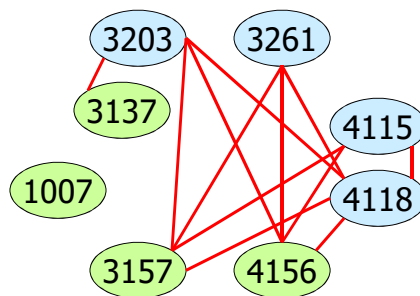
8



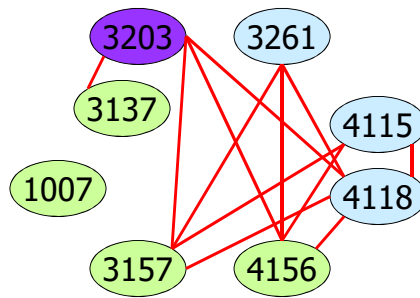
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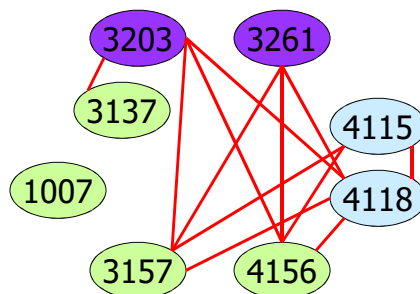
Coloring ...



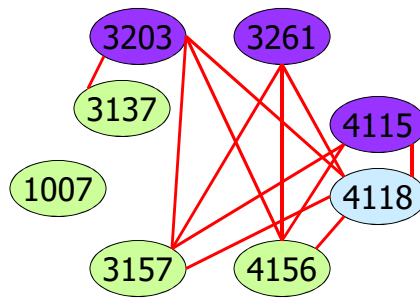
Coloring ...



Coloring ...



Coloring ...



13



2. Applications of graph colorings

(2) Set up natural habitats of animals in a zoo

Solution:

Let the vertices of a graph represent the animals.

Draw an edge between two vertices if the animals they represent cannot be in the same habitat because of their eating habits.

A coloring of this graph gives an assignment of habitats.

14



Homework(part B):

Sec. 10.8 17

